HW1 section 1.1-2, 3, 4, 6, 8

1.1-2 $P(A) = 0.85$ $P(B) = 0.23$
$P(A \cap B) = 0.17$

$P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(A \cap B))$

$= 1 - (0.85 + 0.23 - 0.17) = 0.09$

1.1-3 $P(A) = 12/52$ (Jack, queen, king)
$P(A \cap B) = 2/52$ (Red Jack)

c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(B) = 6/52 \rightarrow P(A \cup B) = 12/52 + 6/52 - 2/52$

d) $P(C \cup D) = 1$ (Club or not club, D is all cards)

e) $P(C \cap D) = 0$

a) $P(A) = 12/52 = 3/13$
b) $P(A \cap B) = 2/52 = 1/26$
c) $P(B) = 6/52 = 3/26$
$P(A \cup B) = 3/13 + 3/26 - 1/26 = 6/26 + 2/26 = 8/26 = 4/13$
d) $P(C \cup D) = 1$
e) $P(C \cap D) = 0$

1.1-4 a) $S = \{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$

b) $A = \{HHHH, HHHT, HHTH, HTHH, THHH$

B = {HHTT, HTHT, HTTH, HTTT, THHT, THTH, THTT, TTHH, TTTH, TTTT}

C = {HHHH, HHHT, HTHH, THHH, TTTT, HTHT, THHT, TTHH, TTHT}

D = {HTTT, THTT, TTHT, TTTH}

i) n(S) = 16 n(A) = 5

P(A) = n(A)/n(S) = 5/16

ii) A ∩ B = {} n(A ∩ B) = 0 P(A ∩ B) = 0

iii) n(B) = 11 P(B) = n(B)/n(S) = 11/16

iv) A ∩ C = {HHHH, HHHT, HTHH, THHH}

n(A ∩ C) = 4 P(A ∩ C) = n(A ∩ C)/n(S) = 4/16

= 1/4

v) n(D) = 4 P(D) = 4/16 = 1/4

vi) P(A ∪ C) = P(A) + P(C) - P(A ∩ C)

n(C) = 8

P(C) = 8/16 = 1/2

∴ P(A ∪ C) = 5/16 + 1/2 - 1/4

= 9/16

vii) B ∩ D = {HTTT, THTT, TTHT, TTTH}

n(B ∩ D) = 4

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$4/16 = 1/4$

1.1-6 $P(A) = 0.5 \quad P(B) = 0.6 \quad P(A \cap B) = 0.4$

a) $P(A \cup B) = 0.5 + 0.6 - 0.40 = 0.70$ (Add rule)

b) $P(A \cap B') = P(A) - P(A \cap B) = 0.5 - 0.4 = 0.1$

c) $P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B) = 1 - 0.4$
$= 0.6$

1.1-8 $P(A' \cap B') = 0.21$

$P(A) = 0.41 \quad P(B) = 0.53$

$P(A \cup B) = 1 - P(A \cup B)' = 1 - P(A' \cap B')$
$= 1 - 0.21 = 0.79$

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.41 + 0.53 - 0.79$

$= 0.94 - 0.79$

$= 0.15$

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